

Name reactions

Aldol condensation

Mannich Reaction

Cannizaro reaction

Rosenmund reduction

Grignard synthesis

Friedel-Crafts alkylation/ acylation

Williamson's ether synthesis

Fisher's esterification

Diazotization reaction

Sandmeyer's reaction

Staudinger's reaction

Wittig reaction

Click reaction

Sonogashira coupling

Suzuki Heck Coupling

Wolf-Kishner reduction

Baeyer-Villiger oxidation

Clemmenson's reduction

Claisen condensation

Michael addition reaction

Diel's alder reaction

Reimer-Tieman reaction

Mitsunobu reaction

Hoffman rearrangement

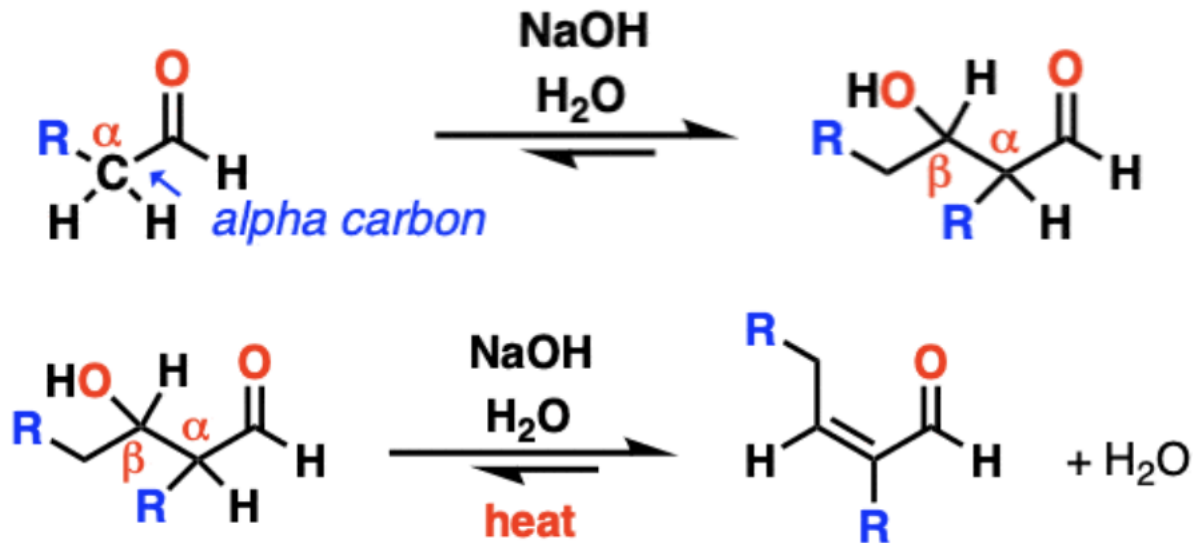
Aldol condensation



Aldehydes and ketones

α,β unsaturated ketones

Self condensation



Why do we require heat?

What temperature?

Base Catalyzed?

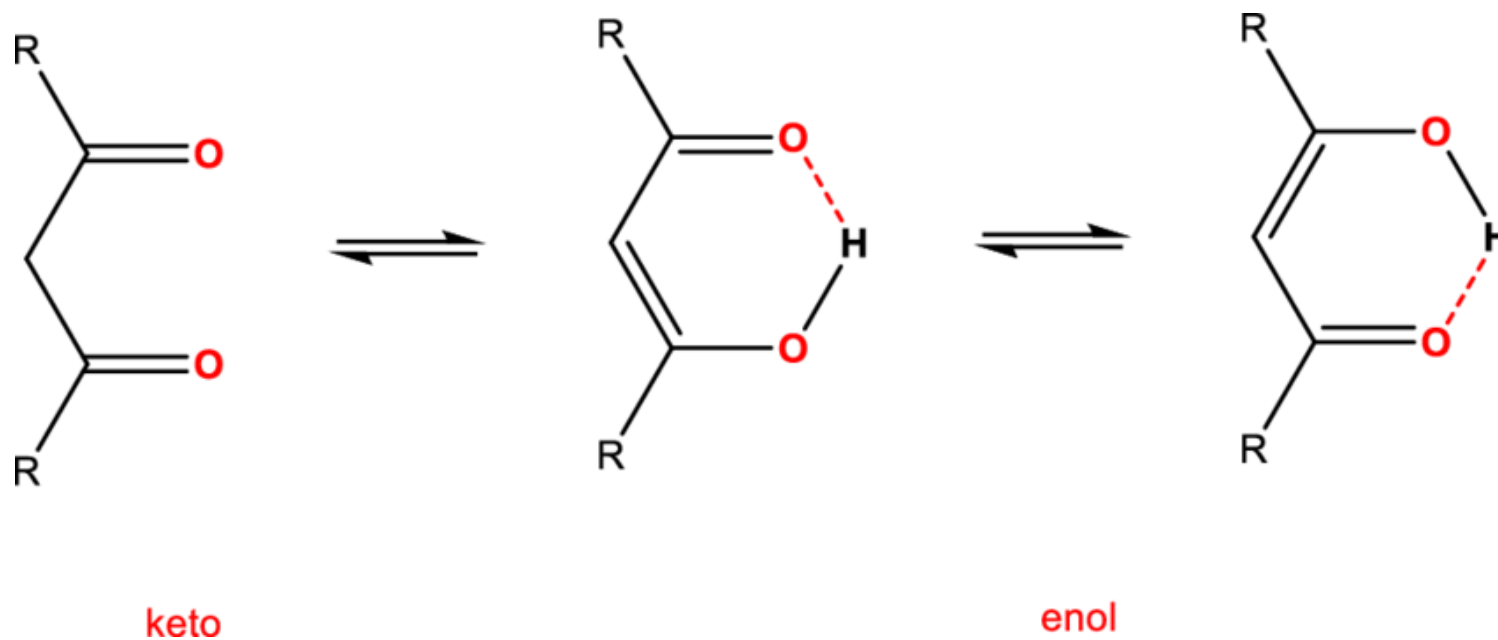
Acid Catalyzed?

Keto-Enol Tautomerism

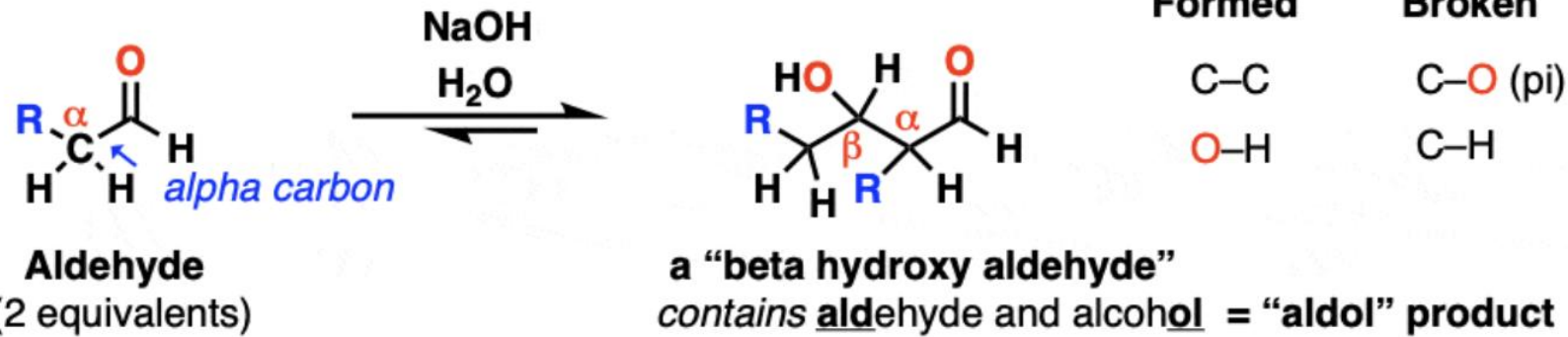


- In general, keto forms are more stable

- In 1,3 diketones, enol form is stabilized via hydrogen bonding



Base-Catalyzed Aldol Reaction ("Aldol Addition")

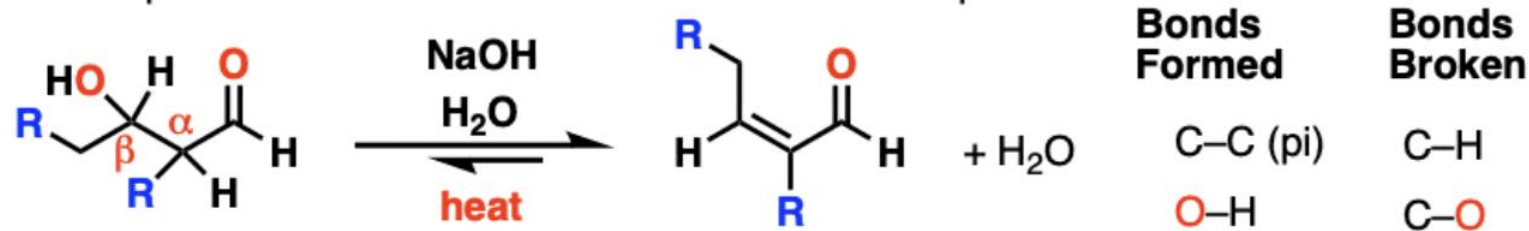


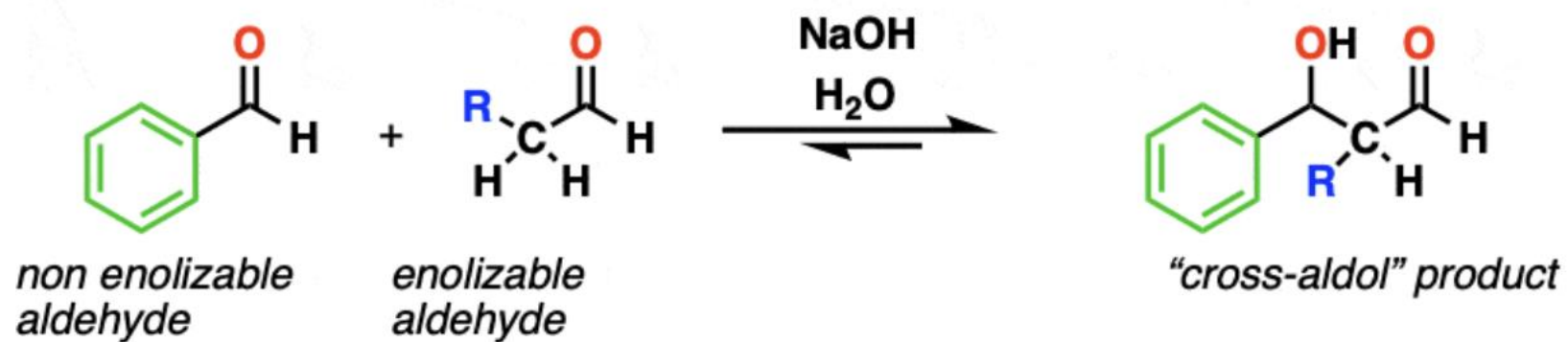
- Base is a catalyst
- Requires aldehyde to have H at the **alpha-carbon**

Mechanism: **D**eprotonation (to form enolate) **A**ddition (of enolate to aldehyde) **P**rotonation

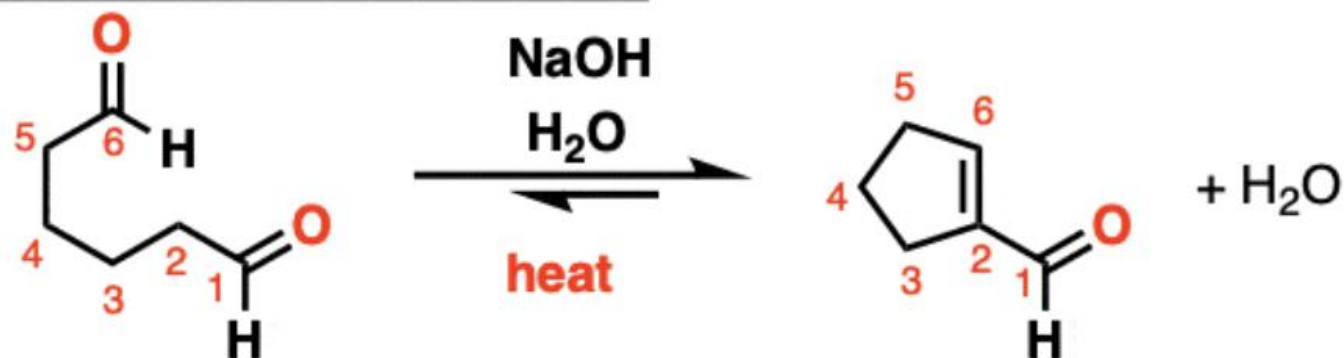
Aldol Condensation

Heating the aldol addition product with base leads to loss of H₂O giving a new C-C pi bond. This product is known as the "aldol condensation" product:



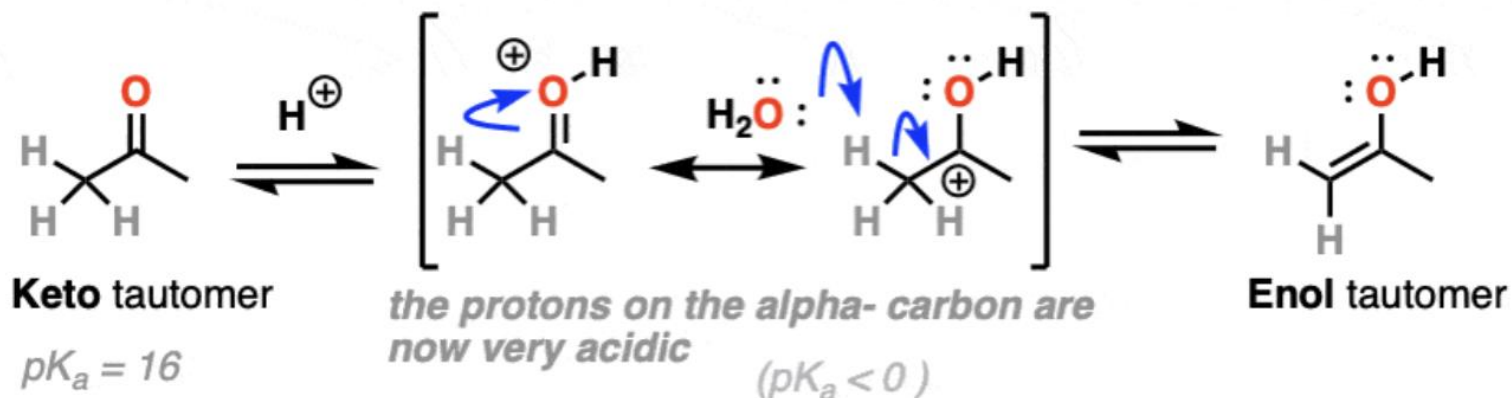


Intramolecular aldol reactions can work well for the formation of 5- and 6-membered rings



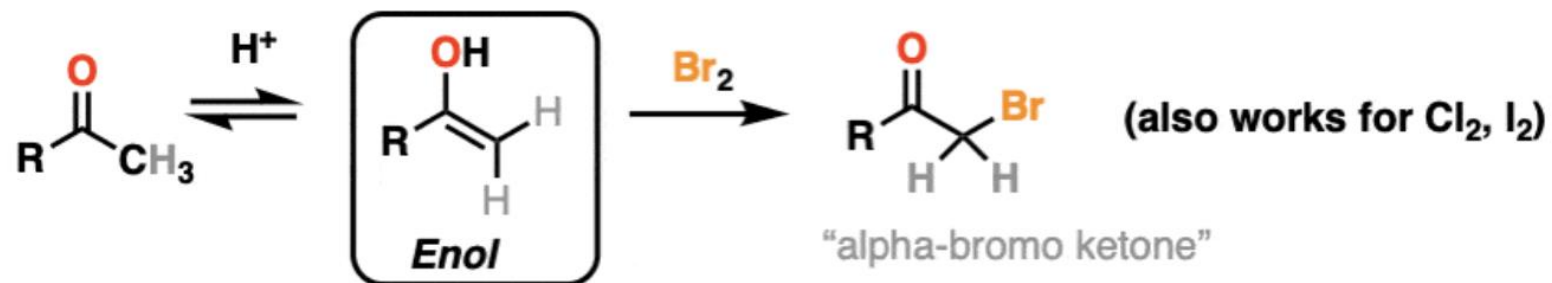
Summary: The Acid-Catalyzed Aldol Reaction (and other reactions of enols)

- Acid catalyzes keto-enol tautomerism (interconversion of keto form and enol form)

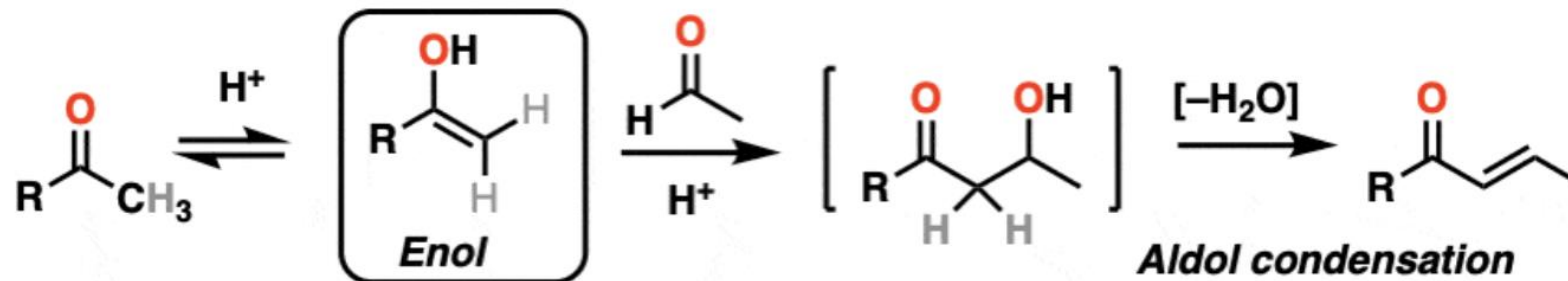


- This is useful because the enol form is a **nucleophile**
- Enols undergo reactions with various electrophiles

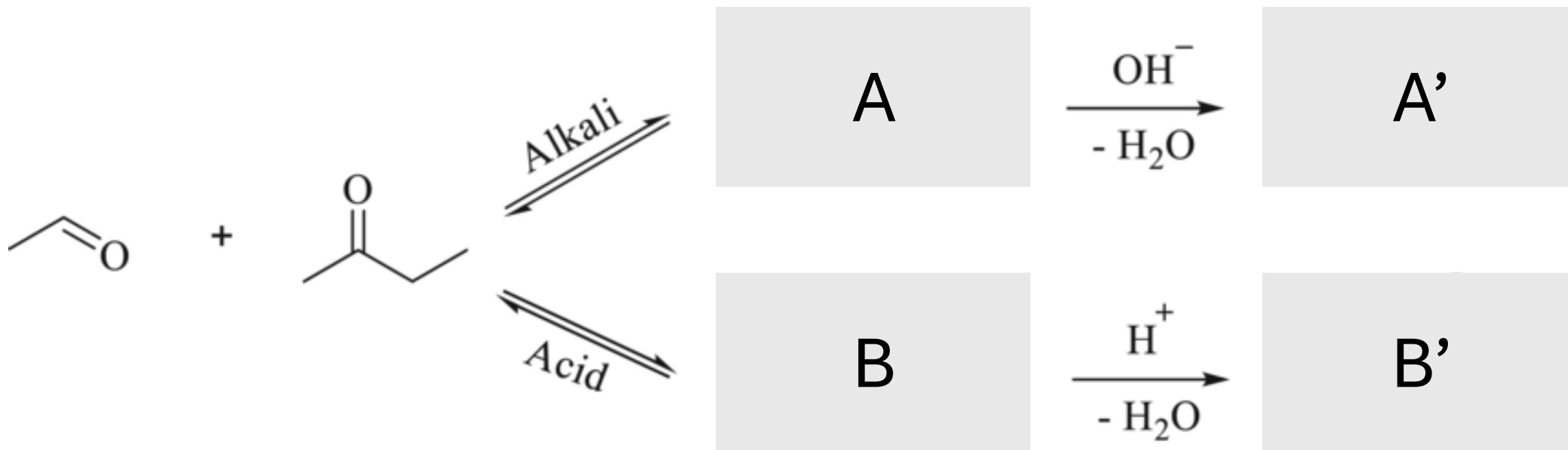
Acid-catalyzed halogenation of ketones



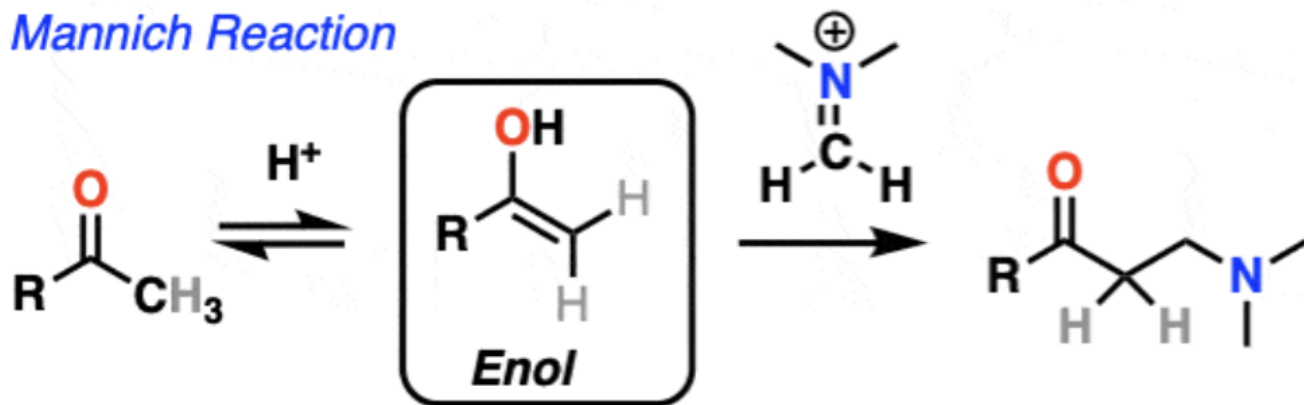
Acid-catalyzed aldol condensation



Predict the structure of the products A, A' and B, B'

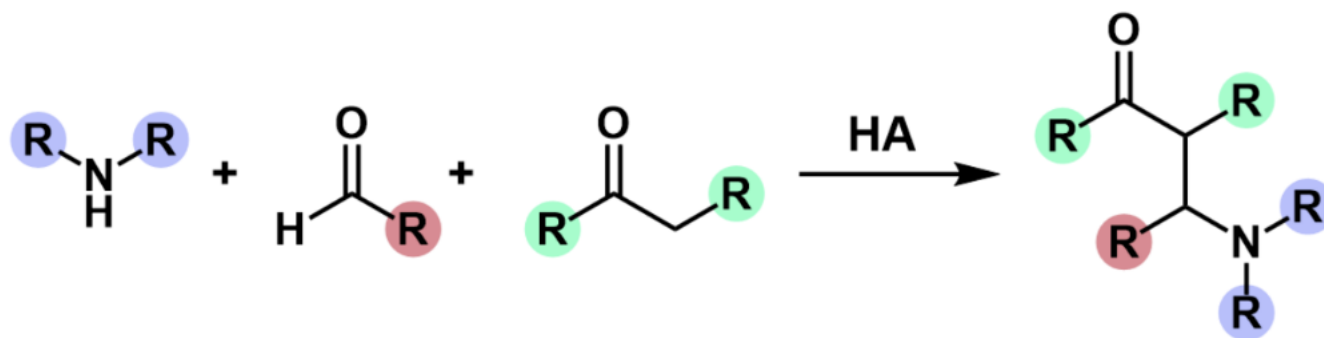


Mannich Reaction



Ketones react with imines to give
 β -amino-carbonyl (Mannich's base)

Aldol + Mannich

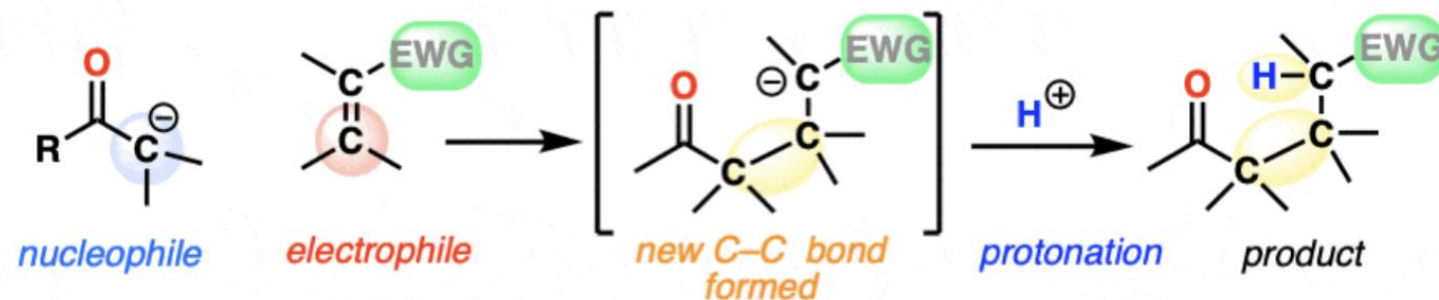


Michael addition reaction

The Michael Reaction (sometimes "Michael Addition")

- addition of an **enolate** (*nucleophile*) to an *electrophilic* alkene conjugated to an electron withdrawing group (EWG)

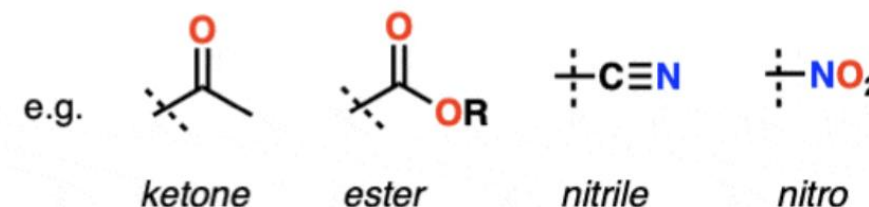
Form	Break
C-C	C-C (pi)
C-H	C-H



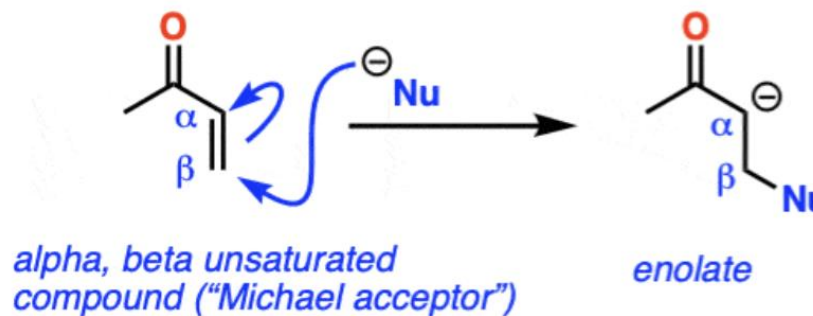
an **enolate**
(usually formed via deprotonation)

an **alpha, beta-unsaturated compound**

EWG = electron-withdrawing group



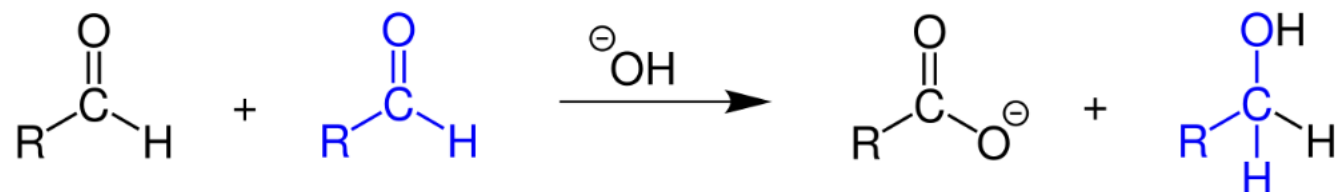
Conjugate addition is the general name for addition of a nucleophile to the beta position of alpha,beta-unsaturated compounds.



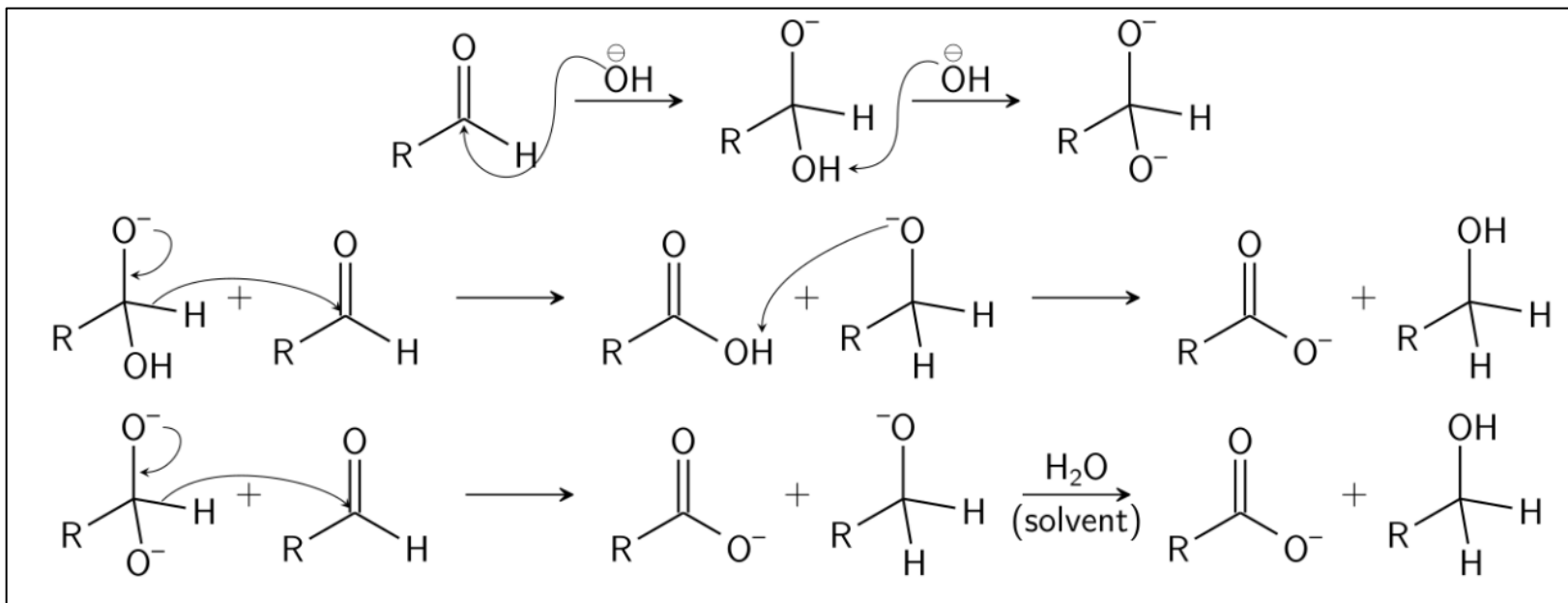
Typical nucleophiles for conjugate addition

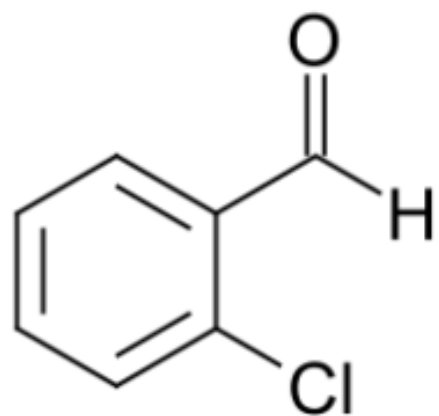
- enolates
- thiols (RSH), thiolates RS^-
- enamines
- amines
- organocuprates (R_2CuLi)

Cannizaro Reaction

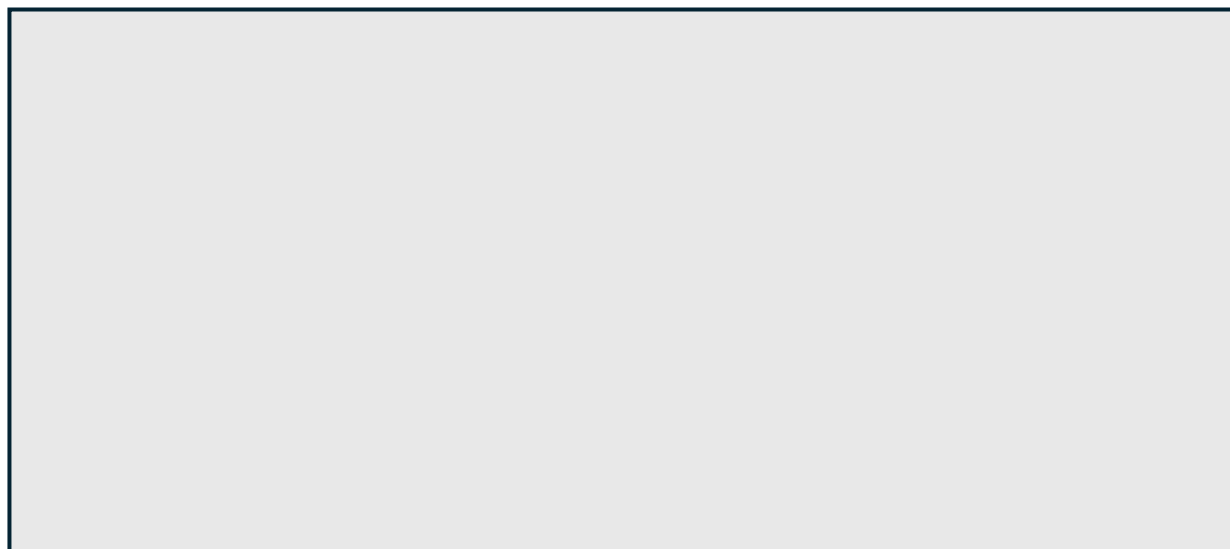


base-induced disproportionation of two molecules of a non-enolizable aldehyde to give a primary alcohol and a carboxylic acid



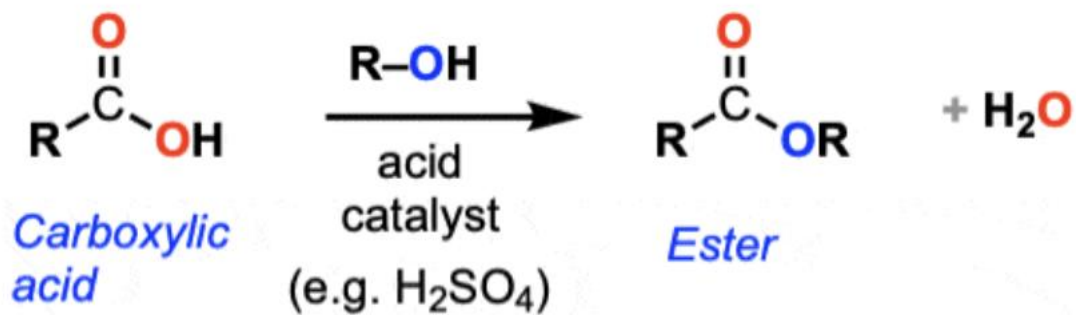


KOH pellets
→
pestle & mortar
30 min.



Fisher Esterification Reaction

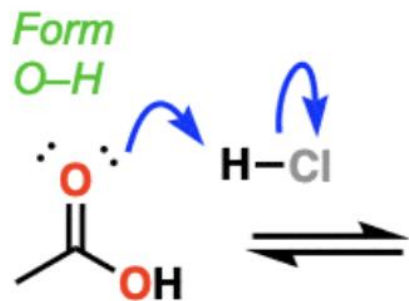
Carboxylic acids to ester



Requires an acid catalyst. Why?

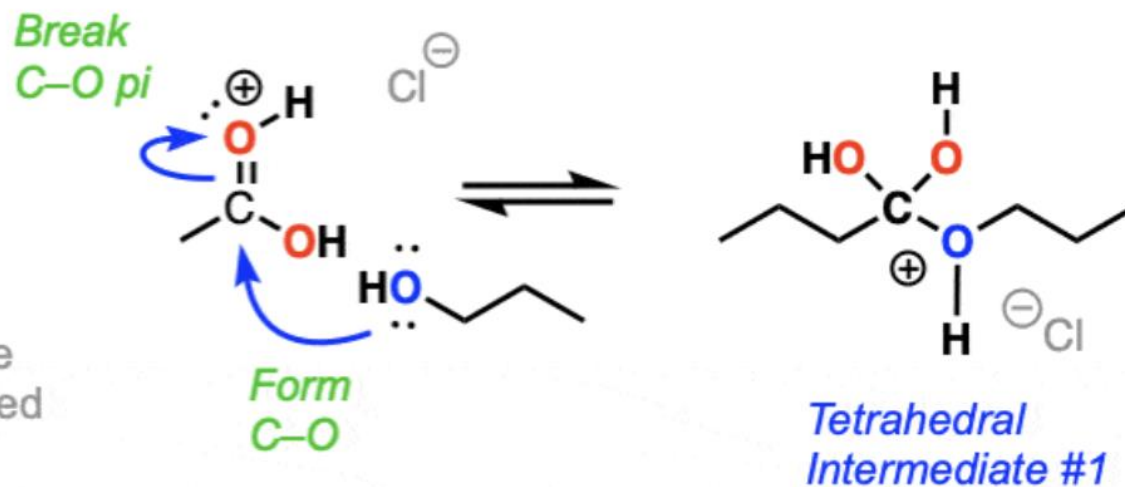
Reaction Mechanism

Step 1: Protonation



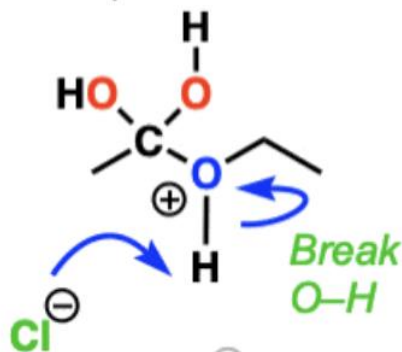
Although H-Cl is shown here many other acids can be used

Step 2: Addition



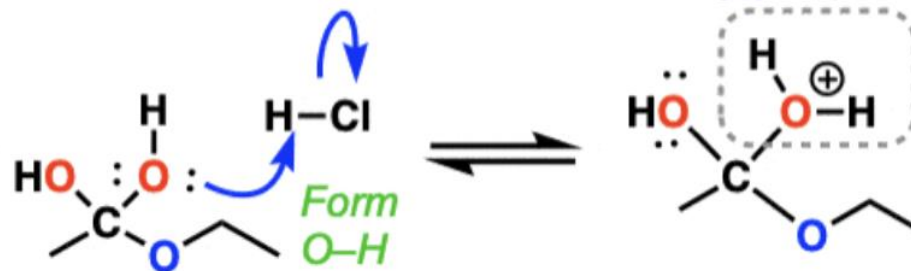
Step 3: Deprotonation

Removal of H-O by a weak base (such as solvent)



Step 4: Protonation

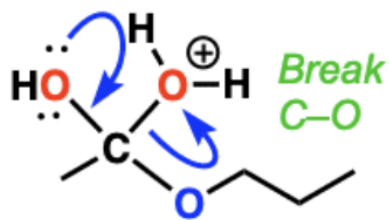
The OH has been converted to the better leaving group OH_2^+



Reaction Mechanism

Step 5: Elimination

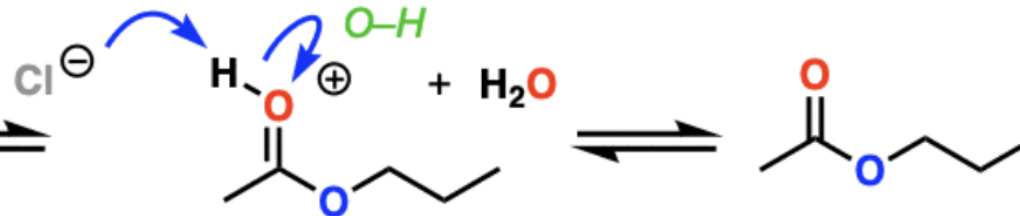
Form
C-O (π)



Step 6: Deprotonation

Any weak base B will work here

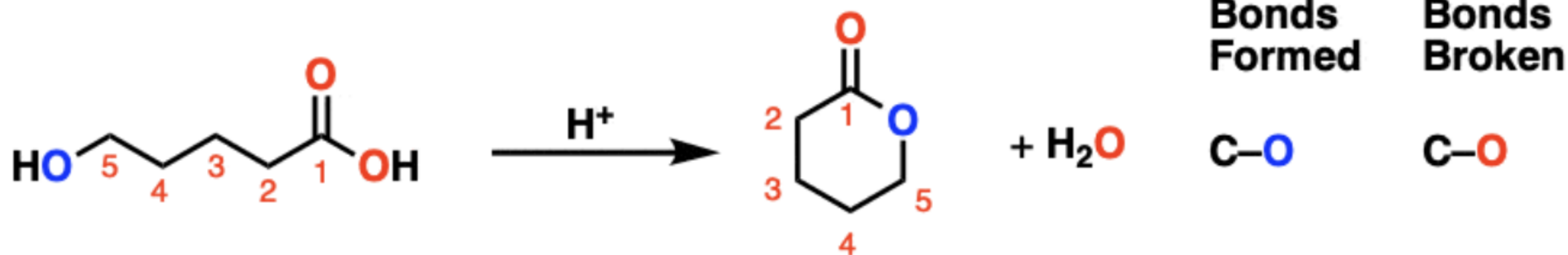
Break
O-H



OH is a weak leaving group

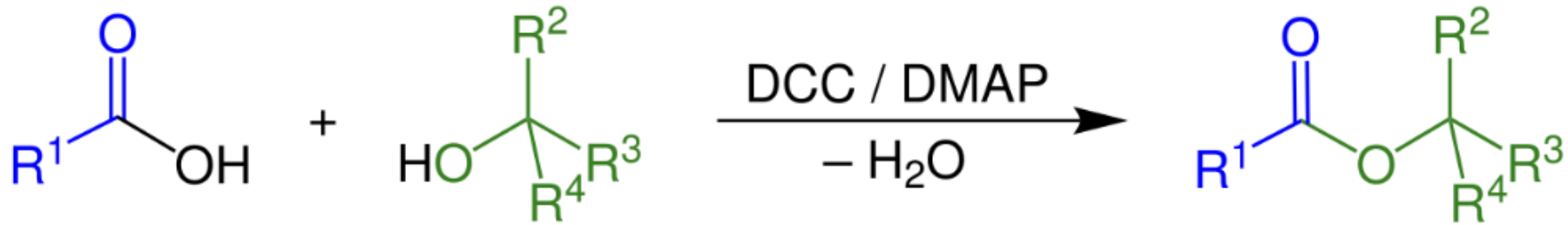
Carboxylic acids require **activation**

Intramolecular Fischer esterification gives a **lactone**



Works best for 5- and 6- membered rings

Steglich esterification



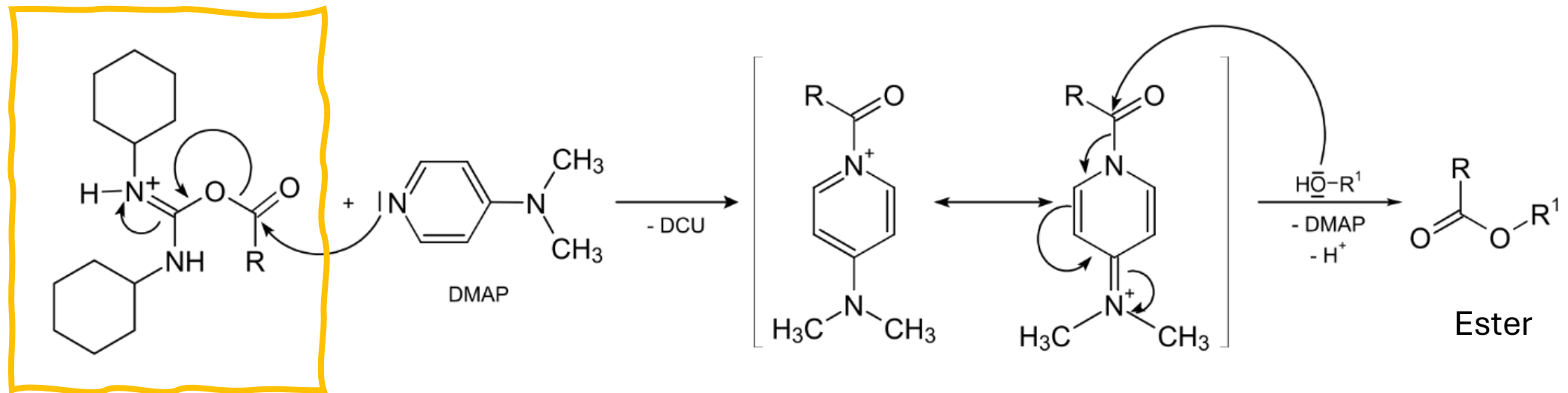
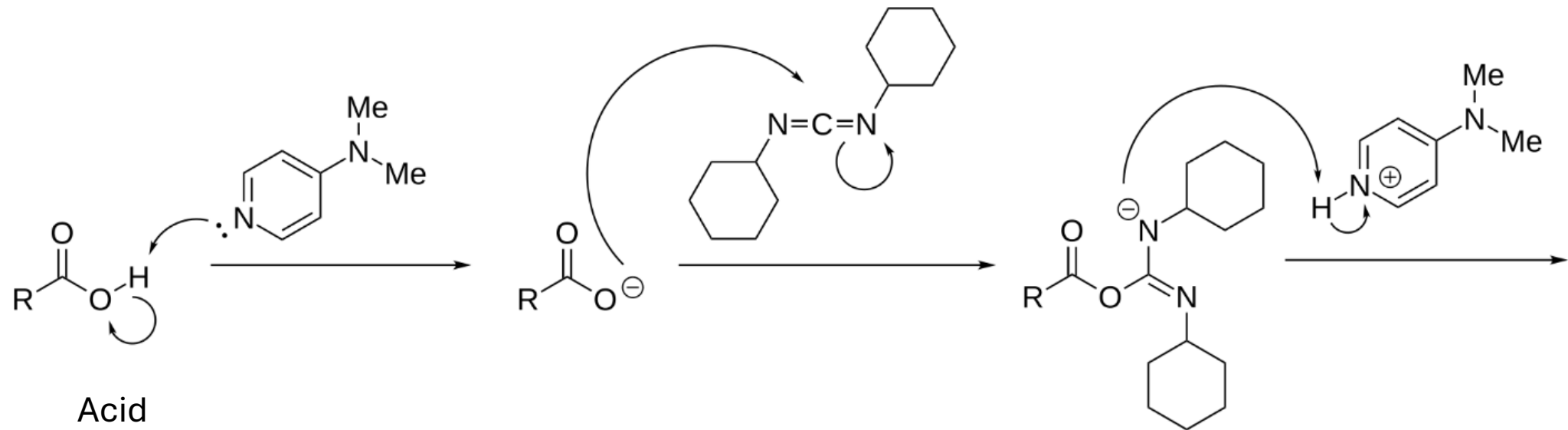
4-dimethylaminopyridine (DMAP)

Dicyclohexylcarbodiimide (DCC)

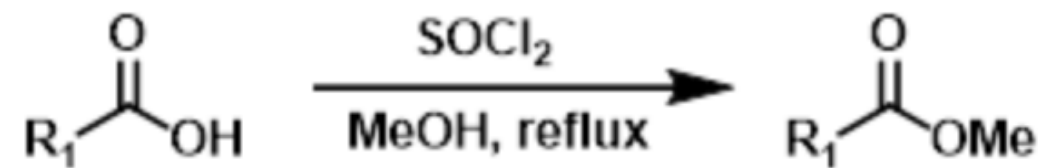
DCC/ DMAP are activating agents for acids

Also used in peptide chemistry (amide bond formation)

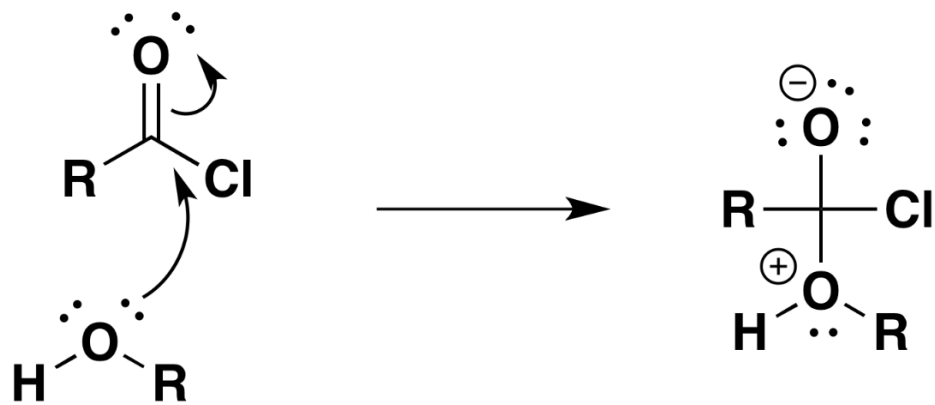
Reaction Mechanism



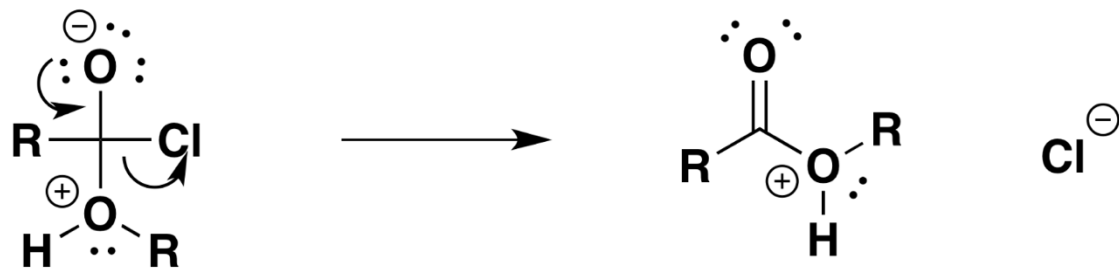
Acid chloride (RCOCl)



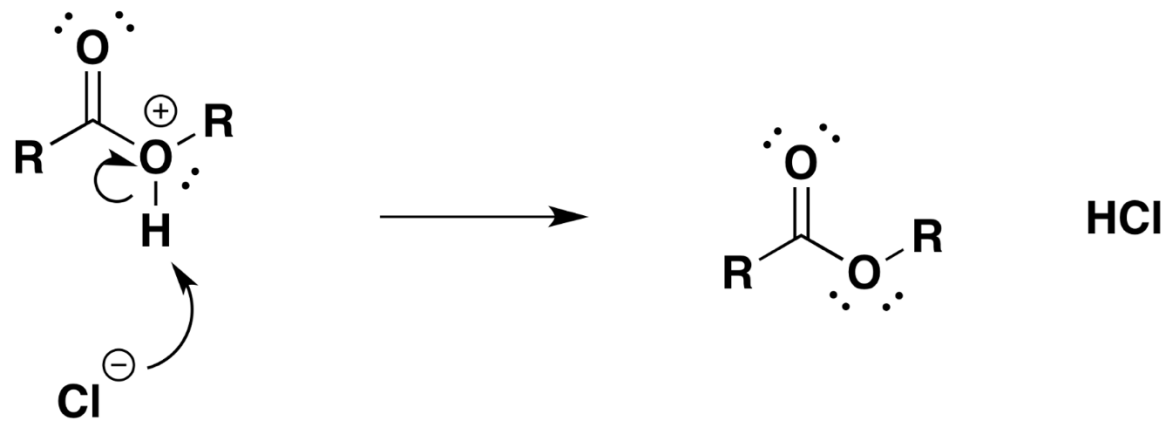
1) Nucleophilic attack by the alcohol



2) Leaving group is removed



3) Deprotonation



Step-by-step
reaction mechanism